

FLOW

1. Landing Page

User website open karta hai. Sabse pehle ek premium landing page dikhti hai jo batati hai ki ye **AI Project Intelligence Platform for Data Centre EPC Projects** hai. Hero section me product ka overview, ek short workflow animation (Upload → AI Analysis → Alerts → Recommendations), key features aur **Get Started / Login** button hota hai.

2. Login

User Login par click karta hai. Usse sirf basic project-related details fill karni hoti hain:

- Company Name
- Work Email
- Password
- Role (Project Manager / Engineer / QA / Admin)
- Project Name (ya existing project select)

Login hote hi user apne selected project ke dashboard par chala jata hai.

3. Main Dashboard (Mission Control)

Ye homepage nahi, balki project ka **Mission Control** hota hai. Yahan user ko ek hi screen par project ki current health dikh jaati hai—Health Score, Progress, Critical Alerts, AI Insights, Recent Activities aur Timeline.

4. Top Navigation Bar (Maximum 4 items)

Main sirf ye 4 cheezein rakhenge:

- **Dashboard** – Overall project overview.
- **AI Project Brain** – AI se interact karna aur AI recommendations dekhna.
- **Global Search** – Documents, vendors, equipment ya tasks search karna.
- **Notifications** – AI alerts aur recent updates.

(Profile icon right corner me rahega.)

5. Right Sidebar

Sidebar me project ke major modules honge:

- **Project Hub** – Project ke documents, schedule, vendors aur equipment ko manage karna.
- **AI Intelligence** – Risk prediction, compliance checking aur what-if simulations dekhna.
- **Commissioning** – Final testing, quality checks aur commissioning progress track karna.
- **Reports** – AI-generated reports aur analytics export karna.
- **Settings** – Users, project aur account preferences manage karna.

Har module ke andar uske related sub-features honge (jaise Project Hub me Documents, Schedule, Vendors, Equipment).

6. User Experience

User chahe to Project Hub me documents upload kare, schedule dekhe ya vendor details manage kare. Background me AI continuously data analyze karta rahega. Agar koi risk, delay ya specification mismatch milta hai to Dashboard aur Notifications me automatically alert aa jayega. User AI Project Brain kholar us risk ka explanation, impact aur recommended solution dekh sakta hai.

Overall Journey

Landing Page → Login → Project Dashboard (Mission Control) → Project Hub me data → AI automatically analyze kare → Dashboard par alerts aur recommendations → User AI Project Brain se details aur decisions dekhe → Reports generate kare.

Member – Role Frontend Developer

User Interface (UI) aur User Experience (UX) banana.

Files/Modules

- **Landing Page** → Platform introduction aur Get Started page.
- **Login Page** → Login form aur user input.
- **Dashboard Page** → Health Score, Alerts, Timeline, AI Recommendations.
- **Project Hub Page** → Documents, Schedule, Vendors, Equipment.
- **AI Project Brain Page** → AI insights aur recommendations dikhane ke liye.
- **Commissioning Page** → Testing aur quality status.
- **Reports Page** → Generated reports.
- **Common Components** → Navbar, Right Sidebar, Cards, Tables, Buttons, Notifications.

Connection: Login page Backend ke Authentication se connect hogi, baaki saare pages Backend ki APIs se data lekar display karenge.

Member – Role Backend Developer

Business Logic, APIs aur Database management.

Files/Modules

- **Authentication Module** → Login aur user verification.
- **Project Module** → Project create, update aur fetch.
- **Document Module** → Document upload aur retrieval.
- **Dashboard Module** → Dashboard ke liye required data provide karega.
- **AI Integration Module** → AI ko request bhejna aur result receive karna.
- **Report Module** → Reports generate aur fetch karna.
- **Database Models** → Users, Projects, Documents, Vendors, Equipment, AI Results ki structure.

Connection: Ye modules Frontend se requests receive karenge, Database se data manage karenge aur AI module se communicate karenge.

Member – Role AI Engineer

Platform ka AI Intelligence develop karna.

Files/Modules

- **Document Processing Module** → PDFs aur documents read karega.
- **RAG Module** → Documents se intelligent answers aur context nikalega.
- **Compliance Engine** → Specification mismatch detect karega.
- **Risk Prediction Engine** → Delays aur risks predict karega.
- **What-If Simulation Engine** → Different scenarios simulate karega.
- **Recommendation Engine** → Best actions suggest karega.

Connection: Backend se documents aur project data receive karega, Database se required information read karega aur analysis ka result Backend ko return karega.

Member – Integration, Testing & Deployment

Sab modules ko connect karna aur final product ready karna.

Files/Modules

- **Integration Module** → Frontend, Backend aur AI ke connections verify karega.
- **Testing Module** → End-to-end testing aur bug tracking.
- **Configuration Module** → Common project settings aur environment setup.
- **Deployment Setup** → Final hosting aur demo-ready configuration.

Connection: Ye ensure karega ki Login se Dashboard tak, Document Upload se AI Analysis tak aur Reports tak poora workflow smoothly chale, bina kisi broken connection ke.

UI Pages & Features

Page / Feature	Description
Landing Page	The entry page of the platform where users understand the purpose of the AI Project Intelligence Platform. It contains a short introduction, platform workflow, key features, product overview, and a Get Started/Login button to begin using the application.
Login Page	Allows users to securely access the platform by entering Company Name, Work Email, Password, Role, and selecting or creating a project. After successful authentication, the user is redirected to the Mission Control Dashboard of the selected project.
Dashboard	Acts as the Mission Control Center of the platform. It displays the overall Project Health Score, project progress, active risks, upcoming milestones, AI insights, critical alerts, AI recommendations, and a recent activity timeline so users can understand the project's current status from a single screen.
Project Hub	The central workspace for managing all project-related information. It contains sections for Documents, Schedule, Vendors, and Equipment where users can upload files, manage project resources, view equipment status, vendor details, and monitor the overall project data used by the AI.
AI Project Brain	A dedicated AI workspace where users can ask project-related questions and receive contextual answers based on uploaded project documents. It also provides AI-generated explanations, recommendations, previous similar cases, and document references for better decision-making.
AI Intelligence	Displays outputs generated by different AI engines, including risk analysis, compliance results, delay predictions, what-if simulations, and recommended actions. Each analysis can be explored in detail to understand why a specific recommendation was generated.
Commissioning	Used during the final project phase to monitor testing and quality assurance. It shows completed, pending, and failed tests, highlights quality issues, and indicates whether the project is ready for final commissioning.
Reports	Provides automatically generated project reports such as daily progress reports, weekly summaries, compliance reports, risk reports, and project status reports. Users can preview, download, or export these reports whenever required.
Global Search	A universal search feature that allows users to search for documents, vendors, equipment, schedules, tasks, or AI recommendations from anywhere in the platform. Selecting a result automatically opens the corresponding section—for example, searching an equipment opens its detailed page inside the Project Hub , where users can view its complete status and related information.
Notifications	Displays important updates generated by the platform, including newly detected risks, vendor delays, uploaded documents, compliance issues, AI recommendations, and project updates. Clicking a notification redirects the user directly to the relevant page or module for further details.
Settings	Allows users to manage their profile, team members, user roles, project information, notification preferences, and other platform configurations required for project management.

Backend & AI Modules

Module / Engine	Description
Document Processing Module	Handles every uploaded PDF, drawing, specification, or project document. It extracts the required content, organizes the information, and prepares the data so that AI modules can efficiently analyze it without requiring manual processing.
RAG Module	Builds a searchable knowledge base from all uploaded project documents. Whenever a user asks a question, it retrieves the most relevant information and provides accurate answers along with references to the original documents for better reliability.
Compliance Engine	Compares project specifications, client requirements, vendor submissions, equipment details, and technical documents to automatically detect mismatches, missing requirements, or non-compliance before they become on-site issues.
Risk Prediction Engine	Continuously analyzes schedules, procurement status, vendor progress, equipment deliveries, and project activities to identify possible delays or risks in advance and estimate their impact on the overall project timeline.
What-If Simulation Engine	Simulates different project scenarios by applying hypothetical changes, such as delayed deliveries or resource shortages, and predicts how those changes will affect milestones, deadlines, costs, and project completion.
Recommendation Engine	Collects outputs from all AI engines, evaluates possible solutions, and generates practical recommendations, recovery plans, alternative vendors, schedule adjustments, or other decision-support suggestions for project managers.
Authentication Module	Manages secure user login, authentication, role verification, session handling, and access permissions to ensure that every user can only access the data allowed for their role.
Project Module	Responsible for creating, updating, selecting, and managing project information, including project metadata, timelines, assigned users, and other project-related records.
Document Module	Handles uploading, storing, organizing, retrieving, and managing project documents. It also ensures uploaded files are available for AI processing whenever required.
Dashboard Module	Collects data from the database and AI modules, processes it, and prepares all information required for the Dashboard, including health score, alerts, progress statistics, milestones, recommendations, and activity timelines.
AI Integration Module	Acts as the communication bridge between the backend and AI services. It sends project data or uploaded documents for AI analysis, receives structured AI responses, stores important results, and forwards them to the frontend.
Database Models	Define the structure for storing Users, Projects, Documents, Vendors, Equipment, Schedules, Notifications, Reports, AI Results, and other project-related data. These models ensure consistent data sharing between the frontend, backend, and AI modules throughout the platform.